

Telecom Customer Churn Analysis and Prediction Using Machine Learning and Customer Segmentation

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1. Introduction / Motivation & Background

Customer churn refers to the situation where customers stop buying a company's products or using its services. With the highly competitive industries like telecommunications today, customer retention has become a key business issue since it is usually more cost-effective to keep existing customers rather than attracting new ones. Besides direct revenue losses, high churn rates may lead to increased marketing expenses and a decrease in customer lifetime value. Hence, it is necessary to know the factors that cause customer churn so as to raise customer satisfaction and create effective customer retention programs. Data analytics and machine learning are rapidly becoming standard tools as companies use customer interaction data to forecast churn risks and pinpoint customer segments with the highest likelihood to churn. Machine learning methods can find complex connections and patterns in customer data, which is a step ahead of traditional descriptive analyses of business performance. The objective of this project is to analyze customer churn behaviors using statistical techniques, machine learning algorithms, and segmentation approaches. It is trying to figure out what the main reasons for customer churn are by using statistical methods, it plans to create and evaluate many different machine learning models based on prediction performance, and it will classify customers according to their behavior as a way to back up business strategies. What this project really wants to do is make churn prediction more accurate but, at the same time, give businesses practical ideas.

Keywords: *Customer Churn, Machine Learning Models, Customer Segmentation Techniques, Predictive Analytics, Classification Models, Feature Importance Analysis, K-means Clustering, PCA, t-SNE Algorithm, Customer Retention*

2. Related Work & Existing Techniques

Customer churn prediction is one of the hottest research topics in industries like telecommunications and subscription-based services, where retaining customers is the key to sustain financially. Most studies, on the one hand, combine traditional statistical methods and on the other hand use advanced machine learning techniques in order to detect churn patterns and boost prediction accuracy (Verbeke et al., 2012). Traditional statistical methods are widely used for identifying the relationship between customer characteristics and their likelihood to churn. For categorical

attributes one can use the Chi-square test of association between variables (Pearson 1900), while for numerical attributes one can use ANOVA test for differences between means of groups (Fisher 1925).

Logistic regression models are quite popular within the machine learning literature on churn prediction due to their interpretability and simplicity. They usually present satisfactory predictions on average as a baseline predictive tool (Hosmer et al. 2013). Advanced ensemble models like Random Forest or Gradient Boosting Machines are commonly employed as they tend to capture non-linear relationships among features leading towards higher predictive power (Breiman 2001; Friedman 2001; Chen and Guestrin 2016).

Customer segmentation is usually employed in order to group customers with similar attributes together. K-Means clustering algorithm is one of most popular algorithms among machine learning practitioners due its simplicity (MacQueen 1967).

Dimensionality reduction techniques like Principal Component Analysis (PCA) or Independent Component Analysis are used regularly in order to visualize and reduce dimensionality of the dataset in question (Jolliffe 2002).

Several recent studies have indicated that integrating statistical modeling supervised machine learning modeling approaches with unsupervised clustering techniques yields strong predictive power combined with interpretability for churn prediction (Verbeke et al., 2012).

3. Methodology

This study uses a combination of statistical analysis, machine learning algorithms development and clustering techniques for analyzing customer churn behavior. The complete project steps include data cleaning and preprocessing, statistical analysis including feature relation analysis with target variable Churn, feature engineering for creating new features related with Business logic of Churn rate calculation feature selection from a list of features through python packages, Handling Imbalance Problem through python package and Machine Learning model making using Logistic Regression model for classifying the classified label onto test data set for checking the accuracy.

3.1 Data Preprocessing

The original dataset contains numerical and categorical customer information. Before the analysis, preprocessing on this dataset was implemented to improve the data quality and model performance.

First, the irrelevant or weakly related features were removed. Such as features having nearly zero correlation with the target variable churn like gender, PhoneService, etc. were removed so that the noise in the data can be reduced.

After removal we converted all the categorical variables into numerical

representations by using the techniques like Label Encoding and One Hot Encoding wherever required. Also, we normalized the numerical variables so as to bring it on same scale.

3.2 Statistical Analysis

To find out significant variables which are affecting the customer churn, we conducted statistical tests on both categorical as well as numerical variables.

For Categorical Variables : Chi-Square Test was conducted to find out relationship between the two categorical variables. Post that cross-tabulation was done to observe and understand the effect of different categories and churn rate.

For Numerical Variables: We used ANOVA Analysis of Variance to compare means of different groups of a variable. So similar way we checked the average for different variables with respect to churn and non-churn.

```
def KF(x):
    df1 = pd.crosstab(df['Churn'], df[x])
    li1 = list(df1.iloc[0, :])
    li2 = list(df1.iloc[1, :])
    kf_data = np.array([li1, li2])

    kf = chi2_contingency(kf_data)
    p_value = kf[1]

    if p_value < 0.05:
        print(
            'Chi-square test p-value for churn by {} is {:.2e}, less than 0.05, indicating significant differences between groups, suitable for [cross-analysis]'
            .format(x, p_value),
            '\n'
        )
    else:
        print(
            'Chi-square test p-value for churn by {} is {:.2e}, greater than 0.05, not suitable for cross-analysis'
            .format(x, p_value),
            '\n'
        )
    )
```

```
cols = [var for var in kf_var if var not in ['MonthlyCharges', 'tenure', 'TotalCharges']]

ncols = 7
nrows = 2

fig, axes = plt.subplots(nrows, ncols, figsize=(5*ncols, 4*nrows))
axes = axes.flatten()

for i, col in enumerate(cols[:nrows * ncols]):

    ct = pd.crosstab(df[col], df['Churn'], normalize=0)

    if 1 in ct.columns:
        ct = ct.sort_values(by=1, ascending=False)

    ct.plot(
        kind='bar',
        stacked=True,
        ax=axes[i],
        color=[ '#2ca02c', '#d62728' ],
        legend=False
    )

    axes[i].set_title(col)
    axes[i].tick_params(axis='x', rotation=30)

for j in range(len(cols), len(axes)):
    axes[j].axis('off')

handles, labels = axes[0].get_legend_handles_labels()
fig.legend(handles, ['No Churn', 'Churn'], loc='upper right')
```

```
def ANOVA(x):
    li_index = list(df['Churn'].value_counts().keys())
    args = []
    for i in li_index:
        args.append(df[df['Churn'] == i][x])

    w, p = levene(*args)

    if p < 0.05:
        print(
            'Warning: p-value for churn by {} is {:.2e}, less than 0.05, indicating unequal variances'
            .format(x, p),
            '\n'
        )
    else:
        f, p_value = f_oneway(*args)
        print(
            'F-value for churn by {} is {:.4f}, p-value is {:.2e}'
            .format(x, f, p_value),
            '\n'
        )

    if p_value < 0.05:
        print('Significant differences in means\n')
    else:
        print('No significant differences\n')
```

3.3 Feature Engineering

Feature Engineering is one of the most important step before proceeding with model building which can lead to dramatic improvement in model performance. It helps to increase model efficiency in learning the patterns among the variables.

We normalized the numerical variables using Standard Scaler and transformed categorical variables into machine-readable format using label encoding and dummy variable generation wherever needed.

```
num_cols = ['tenure', 'MonthlyCharges', 'TotalCharges']

scaler = StandardScaler()

churn_var[num_cols] = scaler.fit_transform(churn_var[num_cols])
```

```

# ===== 1. Standardize missing-like categories =====
churn_var = churn_var.replace({
    'No internet service': 'No',
    'No phone service': 'No'
})
# ===== 2. Binary encoding (Yes/No → 1/0) =====
binary_cols = [
    'Partner', 'Dependents', 'PaperlessBilling',
    'MultipleLines',
    'OnlineSecurity', 'OnlineBackup',
    'DeviceProtection', 'TechSupport',
    'StreamingTV', 'StreamingMovies'
]
for col in binary_cols:
    churn_var[col] = churn_var[col].map({'Yes': 1, 'No': 0})
# ===== 3. Ordinal encoding =====
churn_var['Contract'] = churn_var['Contract'].map({
    'Month-to-month': 0,
    'One year': 1,
    'Two year': 2
})
churn_var['InternetService'] = churn_var['InternetService'].map({
    'No': 0,
    'DSL': 1,
    'Fiber optic': 2
})
# ===== 4. One-hot encoding (nominal variable) =====
churn_var = pd.get_dummies(
    churn_var,
    columns=['PaymentMethod'],
    drop_first=True
)
# ===== 5. Numeric cleanup =====
churn_var['SeniorCitizen'] = churn_var['SeniorCitizen'].astype(int)
# ===== 6. Convert all boolean columns to int =====
bool_cols = churn_var.select_dtypes(include='bool').columns
churn_var[bool_cols] = churn_var[bool_cols].astype(int)

```

3.4 Handling Imbalanced Data

Since it is a case of imbalanced dataset we applied a technique called imblearn before training our model.

We used Stratified Sampling while splitting train-test data which helped us to maintain equal proportion of target class in both training and testing.

Post this we used SMOTE (Synthetic Minority Over-sampling Technique) method to oversample only on training data to handle class imbalance hence making models more sensitive towards churning cases.

```

x = churn_var
y = df['Churn']

sss = StratifiedShuffleSplit(n_splits=5, test_size=0.2, random_state=0)

smote = SMOTE(random_state=0)

for train_index, test_index in sss.split(x, y):
    x_train, x_test = x.iloc[train_index], x.iloc[test_index]
    y_train, y_test = y.iloc[train_index], y.iloc[test_index]

    x_train, y_train = smote.fit_resample(x_train, y_train)

```

3.5 Machine Learning Models

We have built multiple Machine learning algorithms namely Logistic Regression, Naive Bayes, Decision Tree, Random Forest, Support Vector Machine (SVM), AdaBoost, Gradient Boosting and XGBoost CatBoost

```

Classifiers = [
    ['Random Forest', RandomForestClassifier()],
    ['Support Vector Machine', SVC()],
    ['LogisticRegression', LogisticRegression()],
    ['Naive Bayes', GaussianNB()],
    ['Decision Tree', DecisionTreeClassifier()],
    ['AdaBoostClassifier', AdaBoostClassifier()],
    ['GradientBoostingClassifier', GradientBoostingClassifier()],
    ['XGB', XGBClassifier()],
    ['CatBoosting', CatBoostClassifier(logging_level='Silent')]
]

classify_result = []
names = []
predictions = []

for name, classifier in Classifiers:
    classifier.fit(x_train, y_train)
    y_pred = classifier.predict(x_test)

    recall = recall_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred)
    f1score = f1_score(y_test, y_pred)

    class_eva = pd.DataFrame([recall, precision, f1score])
    classify_result.append(class_eva)

    names.append(pd.Series(name))
    predictions.append(pd.Series(y_pred))

```

3.6 Feature Importance Analysis

As Model interpretability is an important aspect we have also done feature importance using various models.

Feature importance is done through different models like Logistic Regression, Naive Bayes, GBDT etc. We combined all feature importances at one place so that feature importance can be looked at better perspective which includes linear relationships, non-linear relationship and so on.

3.7 Customer Segmentation

Further we did clustering of customers using K-Means Clustering on standardized customer attributes.

```
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X_scaled)
data_model["Cluster"] = clusters
```

To visualize this clustering we tried dimensionality reduction technique such as PCA (Principal Component Analysis) which helped us to see how many dimensions are required to explain total variation in the data and t-SNE(t-Distributed Stochastic Neighbor Embedding) which captured relationship between datapoints better by giving different relation between clusters.

```
pca_3d = PCA(n_components=3)
X_pca_3d = pca_3d.fit_transform(X_scaled)

pca_df = pd.DataFrame(X_pca_3d, columns=['PC1', 'PC2', 'PC3'])
pca_df['Cluster'] = data_model['Cluster'].astype(str)

fig = px.scatter_3d(
    pca_df,
    x='PC1',
    y='PC2',
    z='PC3',
    color='Cluster',
    title='PCA 3D Clustering'
)
```

```
tsne = TSNE(n_components=2, perplexity=30, random_state=42)
X_tsne = tsne.fit_transform(X_scaled)

plt.figure(figsize=(8,6))
sns.scatterplot(
    x=X_tsne[:,0],
    y=X_tsne[:,1],
    hue=data_model['Cluster'],
    palette='Set1'
)
```

After this clustering visualisation is done followed by cluster profiling. During Cluster Profiling we compared clusters with respect to churn rates and behavioural characteristics of customers.

4. Dataset & Experimental Results

4.1 Dataset Description

The dataset that has been taken in this project is Telco Customer Churn Dataset In

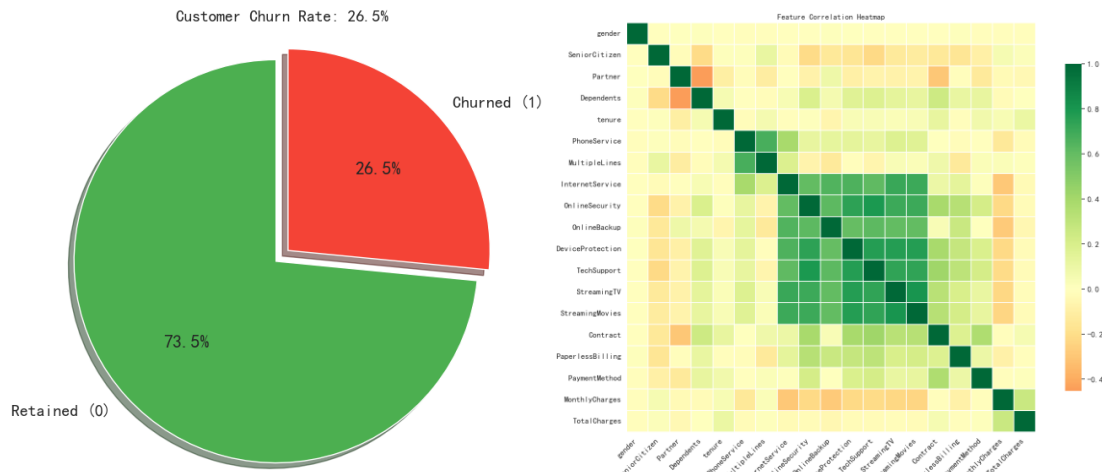
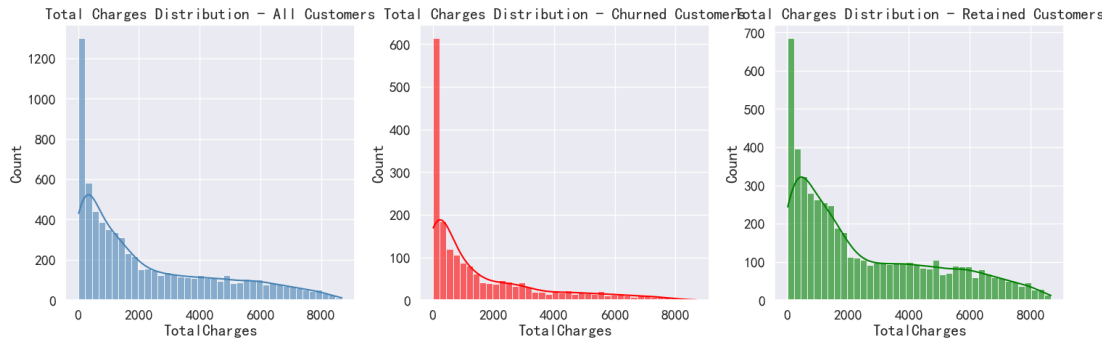
brief we have customer-level information about a telecom company when they have subscribed for their service like demographic information, phone call details, type of contract etc., Finally if the client has been churned or not? From each record it represents customer's information. The dataset includes both numerical and categorical features with target variable Churn indicating whether a client has unsubscribed or not?

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
7595-F38T20	Female	0	Yes	No	1	No	No phone service	DIGL	No	Yes	No	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
5573-GV3TDE	Male	0	No	No	14	Yes	No	DIGL	Yes	No	Yes	No	No	No	One year	No	Mail check	56.95	1889.5	No
1668-GP1BK	Male	0	No	No	2	Yes	No	DIGL	Yes	Yes	No	No	No	No	Month-to-month	Yes	Mail check	51.85	188.15	Yes
1795-CFOCB	Male	0	No	No	45	No	No phone service	DIGL	Yes	No	Yes	Yes	No	No	One year	No	Bank transfer (automatic)	42.5	1840.71	No
9237-BG2TU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes	Electronic check	70.7	121.65	Yes

4.2 Data Exploration Results

While performing initial exploratory analysis on the given data we observed that customer churn is not random distribution across data rather it depends on specific attributes of a customer such as contract, tender value and monthly charges paid by them.

Some categorical variables exhibit strong imbalance in churn distribution, suggesting potential predictive signals for the classification models.



4.3 Model Performance Comparison

Multiple machine learning models have been trained and evaluated using recall,

precision and F1-score. The main results are summarized below:

- Logistic Regression achieved stable performance with good interpretability.
- Ensemble methods such as AdaBoost and Gradient Boosting provided higher overall performance.
- Tree-based models had better ability to capture nonlinear relationships.
- Simple models like Decision Trees didn't perform as well as ensemble models.
- Boosting-based models provided the best tradeoff between precision and recall

	Random Forest	Support Vector Machine	LogisticRegression	Naive Bayes	Decision Tree	AdaBoostClassifier	GradientBoostingClassifier	XGB	CatBoosting
recall	0.655080	0.788770	0.812834	0.810160	0.561497	0.826203	0.799465	0.689840	0.700535
precision	0.571096	0.500849	0.505824	0.523316	0.466667	0.519328	0.532028	0.551282	0.552743
f1score	0.610212	0.612669	0.623590	0.635887	0.509709	0.637771	0.638889	0.612827	0.617925

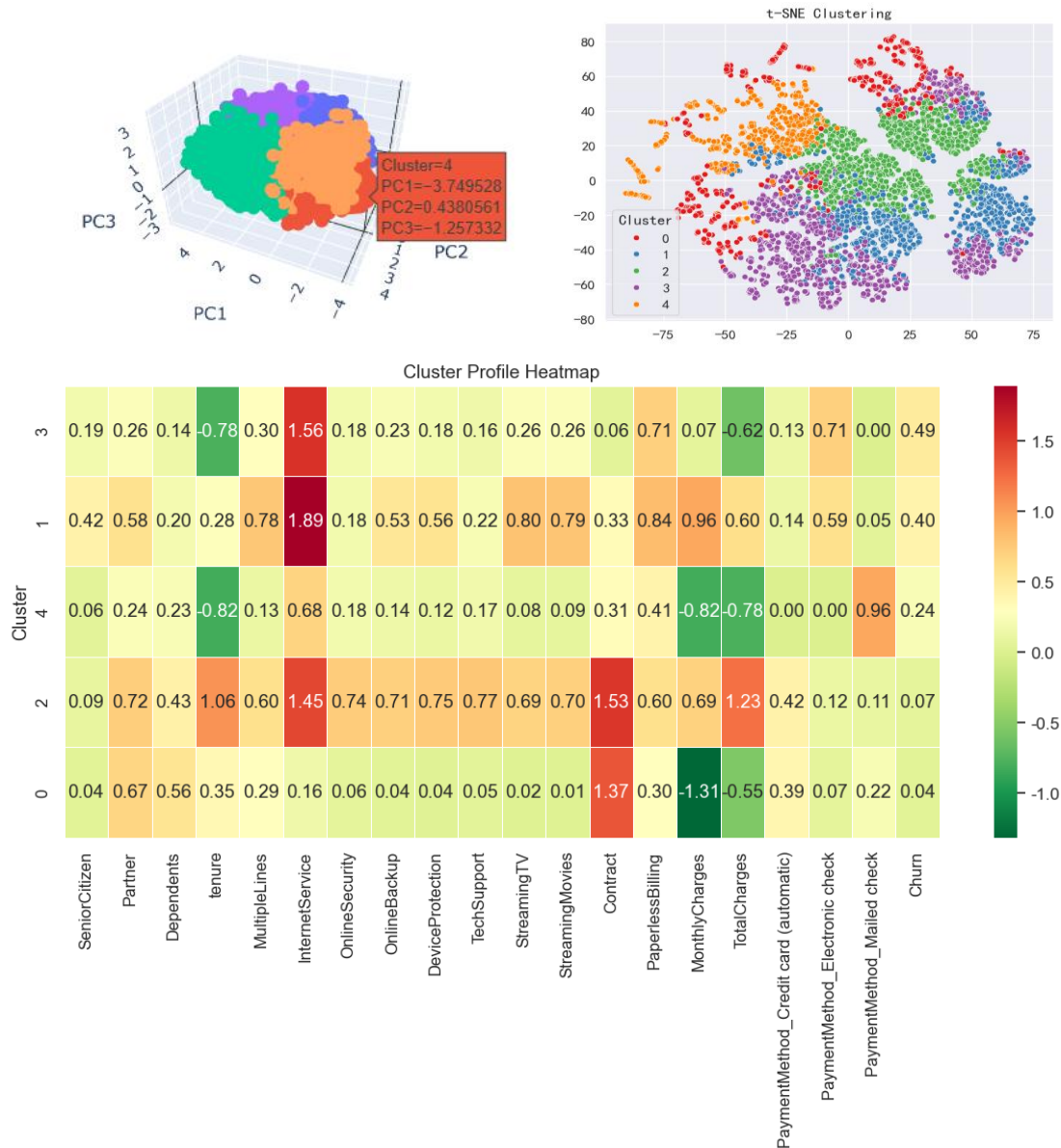
4.4 Feature Importance Results

Feature importance of various models were explored, including Logistic Regression, Naive Bayes, and Gradient Boosting decision trees. The consolidated ranking at the end revealed the most important factors influencing churn are: Monthly charges, tenure (duration of customer account), total charges, and the number of referrals. This implies that contract offerings and pricing incentives are the main factors behind churn behavior.

	Feature	rank_lr	rank_nb	rank_gb	final_rank
0	tenure	1.0	1.0	2.0	1.333333
3	Contract	4.0	2.0	1.0	2.333333
1	TotalCharges	2.0	4.0	6.0	4.000000
4	MonthlyCharges	5.0	5.0	4.0	4.666667
2	OnlineSecurity	3.0	8.0	5.0	5.333333
5	TechSupport	6.0	9.0	7.0	7.333333
18	InternetService	19.0	3.0	3.0	8.333333
9	Dependents	10.0	10.0	9.0	9.666667
6	PaymentMethod_Mailed check	7.0	15.0	8.0	10.000000
7	PaymentMethod_Credit card (automatic)	8.0	12.0	12.0	10.666667
8	OnlineBackup	9.0	13.0	10.0	10.666667
12	Partner	13.0	7.0	14.0	11.333333
16	PaymentMethod_Electronic check	17.0	6.0	11.0	11.333333
14	PaperlessBilling	15.0	11.0	15.0	13.666667
10	DeviceProtection	11.0	14.0	17.0	14.000000
13	SeniorCitizen	14.0	16.0	16.0	15.333333
15	StreamingMovies	16.0	18.0	13.0	15.666667
11	MultipleLines	12.0	19.0	18.0	16.333333
17	StreamingTV	18.0	17.0	19.0	18.000000

4.5 Customer Segmentation Results

K-Means clustering was performed on the dataset to segment customers into different clusters. 5 segments were identified using two dimensionality reduction techniques – PCA and t-SNE. It was found customers form a continuous distribution rather than clearly separate clusters. The distribution is uniform across tenure but is concentrated more towards lower monthly charges for customers who churn.



5. Result Analysis & Conclusion

This paper integrates statistical evaluation, machine learning model development and customer segmentation to map a detailed picture of customer attrition. Findings reveal that customer attrition is influenced by behavior and service factors rather than occurring randomly. The main factors leading to customer attrition, confirmed through various analytical methods - statistical tests, machine learning algorithms, or clustering methods - include: contract type, length of time with the company, and monthly charges. Customers on short-term contracts with higher monthly fees were more likely to leave. Also, a Chi-squared analysis unveiled that churn has statistically significant associations with other categorical variables. Once more, length of time with the company and contract type emerge as the two most important features for prediction in a machine learning model based on feature importance analysis. Segmentation findings demonstrate that the probability of churn differs among

customer groups instead of being the same for all. From a managerial perspective, it's clear that customer retention programs should be focused on high-risk group at least if not all groups. High-risk group which accounts for nearly 20% of total subscribers are contributing around 50% of revenue. So targeting this group can have significant impact on sales revenue. This can be done by providing discounts or offers for long term contracts so that they don't leave any time soon.

6. Future Work

Although satisfactory performance has been achieved in terms of prediction power and business insight derivation through our analysis several extensions could be considered.

- Instead of simple averaging you could use stacking or weighted voting to combine different model predictions.
- Threshold can be tuned depending upon priorities in terms of recall and precision in high-risk customer detection.
- SHAP values can be used to make feature attribution more consistent across all models.
- Adding temporal information in terms of changes in features over time could potentially give you a significant boost for prediction power for churn analysis
- Using deep learning unsupervised techniques to identify complex relationships among clients behaviors may reveal new insights for churn management

7. References

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